

LoanEval AI

Multi-Step Prompt Workflow for Loan Application Evaluation

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Assessment Type	Mini Assessment – Prompt Workflow Design

1. Project Objective

LoanEval AI is a structured, multi-step prompt workflow that takes a loan application and supporting documents as input, evaluates the information from multiple perspectives, identifies inconsistencies and missing information, and produces a concise review-ready assessment. Each stage passes its output as context to the next stage.

2. Problem Statement

Loan applications often contain information across forms, identity documents, income records and bank statements. Manual review can make it difficult to identify missing information and inconsistencies systematically. LoanEval AI divides the evaluation into six connected stages.

3. Workflow Architecture

1 Application Understanding	2 Document Analysis	3 Consistency Evaluation	4 Risk & Impact Assessment	5 Application Optimization	6 Final Assessment
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4. Workflow Steps and Expected Outputs

STEP	PURPOSE	EXPECTED OUTPUT
1. Application Understanding	Understand the loan request, applicant profile, income, employment, obligations and requirements.	Structured application summary with key requirements and constraints.
2. Document Analysis	Extract relevant information from supporting documents and identify missing or unclear fields.	Document findings, extracted fields and missing information.
3. Consistency Evaluation	Compare application information with supporting documents and identify mismatches.	Consistency report with matches, mismatches and verification items.
4. Risk & Impact Assessment	Evaluate information quality, review risks, complexity and available affordability indicators.	Scorecard with risks, explanations and mitigation actions.
5. Application Optimization	Improve completeness, clarity, verification and reviewer usability.	Improved checklist, missing documents and recommended actions.
6. Final Assessment	Combine validated findings into a concise reviewer-ready assessment.	Final summary, findings, score, risks and required actions.

5. Detailed Prompt Design

Step 1 — Application Understanding

Analyze the provided loan application. Identify the loan purpose, requested amount, applicant profile, employment, income, existing obligations, constraints and information needed for a complete review. Do not make an approval or rejection decision at this stage.

Step 2 — Supporting Document Analysis

Using the application context, analyze the supporting documents. Extract relevant information such as identity, employment, income, dates and account information. Identify missing, unreadable or ambiguous fields. Do not assume information that is not present.

Step 3 — Consistency & Uniqueness Evaluation

Compare the application information with extracted document information. Identify matches, mismatches, missing values and unclear discrepancies. Explain which fields require human verification. Provide a consistency score from 1–10 with justification.

Step 4 — Risk & Assessment Review

Evaluate documentation completeness, income consistency, obligation visibility, information quality, affordability-related indicators and review complexity. Score each category from 1–10, calculate an overall assessment score, and identify three major review risks with one mitigation action for each.

Step 5 — Application Optimization

Improve the application workflow using previous findings. Preserve the applicant's original request while strengthening completeness, clarity, verification and reviewer usability. Provide an improved summary, five recommended actions and missing-document requirements.

Step 6 — Final Loan Assessment Generator

Using all previous analyses, generate the final reviewer-ready assessment. Include application summary, loan purpose, requested amount, document findings, consistency findings, assessment scores, risks, verification actions, missing documents and an overall evaluation score. Do not invent facts or financial records.

6. Prompt Chaining & Information Flow

Step 1 creates the application analysis; Step 2 consumes it to analyze documents; Step 3 compares document findings with the application; Step 4 assesses risks using validated information; Step 5 uses the findings to improve the workflow; and Step 6 consolidates the results into a final assessment. This makes the workflow reproducible and traceable.

7. Demonstration Example

Input: An applicant requests a personal loan of \$5,00,000 and provides employment, monthly income and loan-purpose information together with identity proof, a salary slip and a bank statement.

Expected Result: LoanEval AI extracts fields, compares information across sources, highlights inconsistencies, identifies missing information and prepares a structured review summary.

8. Expected Learning Outcome

The workflow demonstrates sequential prompting, contextual reuse, structured outputs, evaluation criteria, iterative improvement and final content generation.

9. Suggested Input Fields

A practical implementation can accept structured application fields together with uploaded supporting documents.

1. Applicant Name	2. Date of Birth
3. Contact Number	4. Email Address
5. Residential Address	6. Employment Type
7. Employer Name	8. Job Title
9. Monthly Income	10. Existing Monthly Obligations
11. Loan Type	12. Requested Loan Amount
13. Loan Tenure	14. Loan Purpose
15. Bank Account Details	16. Identity Document Type
17. Identity Document Number	18. Salary Slip Date
19. Bank Statement Period	20. Supporting Documents Provided

10. Sample Final Output Structure

SECTION	CONTENT
Application Summary	Applicant profile, loan purpose and requested amount.
Document Findings	Documents received, extracted fields and missing information.
Consistency Findings	Matched fields, mismatches and verification requirements.
Assessment Score	Category scores and overall evaluation score.
Key Risks	Three major review risks with mitigation or verification actions.
Required Actions	Corrections, clarifications and additional documents needed.
Final Review Summary	Concise explanation for a human reviewer; not an automatic lending decision.

11. Conclusion

LoanEval AI transforms a loan application into a structured, evaluated and review-ready assessment through six connected prompt stages. The workflow separates application understanding, document analysis, consistency evaluation, risk assessment, optimization and final communication. It is intended to support human review rather than make an automatic lending decision.